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To:
Subject: p and s do not lose

Genome

Further information: Neanderthal genome project

Early investigations concentrated on mitochondrial DNA (mtDNA), which, owing to strictly matrilineal inheritance and subsequent vulnerability to [genetic drift](#), is of limited value in evaluating the possibility of interbreeding of Neanderthals with [Cro-Magnon](#) people.

In 1997, geneticists were able to extract a short sequence of DNA from Neanderthal bones from 30,000 years ago.^[58] The extraction of mtDNA from a second specimen was reported in 2000, and showed no sign of modern human descent from Neanderthals.^[35]



Scientist at the Max Planck Institute extracting the DNA.

In July 2006, the [Institute for Evolutionary Anthropology and Life Sciences](#) announced that they would [sequence the Neanderthal genome](#) over the next two years. This [genome](#) is expected to be roughly the size of the [human genome](#), three-billion base pairs, and share most of its [genes](#). It was hoped the comparison would expand understanding of Neanderthals, as well as the evolution of humans and human brains.^[59]

[Svante Pääbo](#) has tested more than 70 Neanderthal specimens. Preliminary DNA sequencing from a 38,000-year-old bone fragment of a femur found at [Vindija Cave](#), Croatia, in 1980 showed Neanderthals and modern humans share about 99.5% of their DNA. From mtDNA analysis estimates, the two species shared a common ancestor about 500,000 years ago. An article^[60] appearing in the journal [Nature](#) has calculated the species

diverged about 516,000 years ago, whereas fossil records show a time of about 400,000 years ago.^[61] A 2007 study pushes the point of divergence back to around 800,000 years ago.^[62]

Edward Rubin of the Lawrence Berkeley National Laboratory states recent genome testing of Neanderthals suggests human and Neanderthal DNA are some 99.5% to nearly 99.9% identical.^{[63][64]}

On 16 November 2006, Lawrence Berkeley National Laboratory issued a press release suggesting Neanderthals and ancient humans probably did not interbreed.^[65] ████████ director of the U.S. Department of Energy's Lawrence Berkeley National Laboratory and the Joint Genome Institute (JGI), sequenced a fraction (0.00002) of genomic nuclear DNA (nDNA) from a 38,000-year-old Vindija Neanderthal femur. They calculated the common ancestor to be about 353,000 years ago, and a complete separation of the ancestors of the species about 188,000 years ago.^[66]

Their results show the genomes of modern humans and Neanderthals are at least 99.5% identical, but despite this genetic similarity, and despite the two species having coexisted in the same geographic region for thousands of years, Rubin and his team did not find any evidence of any significant crossbreeding between the two. Rubin said, "While unable to definitively conclude that interbreeding between the two species of humans did not occur, analysis of the nuclear DNA from the Neanderthal suggests the low likelihood of it having occurred at any appreciable level."^[66]

In 2008 Richard E. Green et al. from Max Planck Institute for Evolutionary Anthropology in Leipzig, Germany published the full sequence of Neanderthal mitochondrial DNA (mtDNA) and suggested "Neanderthals had a long-term effective population size smaller than that of modern humans."^[67] Writing in *Nature* about Green et al.'s findings, James Morgan asserted the mtDNA sequence contained clues that Neanderthals lived in "small and isolated populations, and probably did not interbreed with their human neighbours."^{[68][69]}

In the same publication, it was disclosed by ████████ that in the previous work at the Max Planck Institute that "Contamination was indeed an issue," and they eventually realized that 11% of their sample was modern human DNA.^{[70][71]} Since then, more of the preparation work has been done in clean areas and 4-base pair 'tags' have been added to the DNA as soon as it is extracted so the Neanderthal DNA can be identified.

With 3 billion nucleotides sequenced, analysis of about 1/3 showed no sign of admixture between modern humans and Neanderthals, according to Pääbo. This concurred with the work of Noonan from two years earlier. The variant of *microcephalin* common outside Africa, which was suggested to be of Neanderthal origin and responsible for rapid brain growth in humans, was not found in Neanderthals. Nor was the *MAPT* variant, a very old variant found primarily in Europeans.^[70]

However, an analysis of a first draft of the Neanderthal genome by the same team released in May 2010 indicates interbreeding may have occurred.^{[13][14]} "Those of us who live outside Africa carry a little Neanderthal DNA in us," said Pääbo, who led the study. "The proportion of Neanderthal-inherited genetic material is about 1 to 4 percent. It is a small but very real proportion of ancestry in non-Africans today," says Dr. David Reich of Harvard Medical School, who worked on the study. This research compared the genome of the Neanderthals to five modern humans from China, France, sub-Saharan Africa, and Papua New Guinea. The

finding is that about 1 to 4 percent of the genes of the non-Africans came from Neanderthals, compared to the baseline defined by the two Africans.^[13]

This indicates a gene flow from Neanderthals to modern humans, i.e., interbreeding between the two populations. Since the three non-African genomes show a similar proportion of Neanderthal sequences, the interbreeding must have occurred early in the migration of modern humans out of Africa, perhaps in the Middle East. No evidence for gene flow in the direction from modern humans to Neanderthals was found. The latter result would not be unexpected if contact occurred between a small colonizing population of modern humans and a much larger resident population of Neanderthals. A very limited amount of interbreeding could explain the findings, if it occurred early enough in the colonization process.^[13]

While interbreeding is viewed as the most parsimonious interpretation of the genetic discoveries, the authors point out they cannot conclusively rule out an alternative scenario, in which the source population of non-African modern humans was already more closely related to Neanderthals than other Africans were, due to ancient genetic divisions within Africa.^[13]

Among the genes shown to differ between present-day humans and Neanderthals were *RPTN*, *SPAG17*, *CAN15*, *TTF1* and *PCD16*.^[13]

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